HACKIIITD 2026 · ROUND 1 · OPEN INNOVATION

EpiRoute

Outbreak intelligence that connects live public-health reports to the air-travel network — and shows the probability, the assumptions and the source behind every number.

- Importation risk
- Local establishment risk
- Sourced outbreak reports

Map: Natural Earth boundaries; arcs are illustrative, not real traffic.

Team details

Team name
[Team name]

[Member 1 name] — Team lead
[Role] · [College, year]
[email] · [phone]

[Member 2 name]
[Role] · [College, year]
[email] · [phone]

[Member 3 name]
[Role] · [College, year]
[email] · [phone]

[Member 4 name]
[Role] · [College, year]
[email] · [phone]

Team size 2–4 · edit or delete unused member rows

Outbreak reports and travel data never meet on one screen

Problem statement

1 Reports are prose, not data

WHO Disease Outbreak News has a public API, but case counts, places and dates often sit inside narrative HTML (we checked: its Epidemiology field is frequently empty). Regional and national sources often have no API at all.

2 Maps show where, not who is next

Disease spreads along flight networks. In Brockmann & Helbing's simulated pandemic, network ‘effective distance’ explained arrival times with R^2 0.97 vs 0.34 for geographic distance.

3 Risk is quoted as one opaque score

Whether an infected traveller lands and whether an outbreak takes off are different events with different inputs. Merging them hides the assumptions people need to judge them.

Target users

Port & airport health officers, district surveillance teams

Which incoming routes deserve screening attention this fortnight?

Travel-medicine clinics, university & corporate travel desks

What is the sourced situation at a destination, and how fast could it change?

Journalists, researchers, students

A transparent, citable explanation of importation risk instead of a black-box score.

Situational awareness only — not diagnosis, not an official forecast, not individual medical advice. Source: Brockmann D, Helbing D. Science 2013;342:1337 (simulation result). WHO DON API checked 15 Oct 2026.

One map: sourced outbreaks, flight links, separate risks



- Pick a reported outbreak
- See linked routes and hubs
- Select a destination country
- Move assumptions, watch ranges shift

Three probabilities, three different inputs — never one score

Already here?

$$P(\text{present})$$

Local case reports + an under-reporting band. Flights are NOT an input — a flight link never implies presence.

Will an infected traveller land?

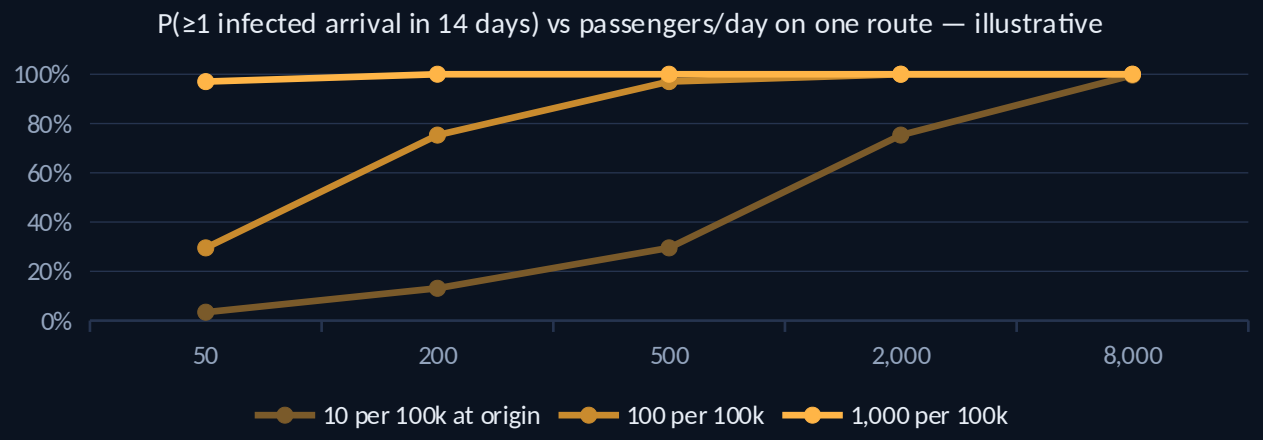
$$P(\text{import, T days}) = 1 - e^{(-\lambda T)}$$
$$\lambda = \Sigma_m \, (I_m/N_m) \cdot F_{mn} \cdot \alpha \cdot (1 - \sigma)$$

I_m/N_m origin prevalence · F passengers/day · α chance an infectious person still boards · σ screening

Will it take off locally?

$$P(\text{establish} \mid 1 \text{ intro}) = 1 - q$$
$$q = (1 + R(1 - q)/k)^{(-k)}$$

Branching process: R reproduction number, k dispersion (cluster-size studies). Vectors / climate decide this for Zika-like pathogens.



Why busy routes ≠ outbreaks

14% of single introductions establish when R = 2.5, k = 0.1 (COVID-like dispersion). Most chains die out.

Ranking who is next: edge $d(m,n) = 1 - \ln(F_{mn} / F_m)$; shortest path over hubs gives effective distance, so Istanbul, Dubai or Singapore transfers count.

Chart: our computation of the Poisson model with α = 0.5, σ = 0 (scenario, not an estimate for any outbreak). Method: briefing note 'How disease-spread probabilities get their data' (14 Oct 2026); Brockmann & Helbing, Science 2013; Lloyd-Smith et al., Nature 2005.

How do you plan to integrate the TinyFish API?

TinyFish turns official web pages into validated, sourced records — and fills the model input no free API provides: this week's border-screening policy (σ).

1 Discover

Search API

include_domains: who.int, ecdc.europa.eu, paho.org, africacdc.org · recency filter

free

2 Watch

Monitor

Page monitor on WHO DON + ECDC threat-report listings → webhook triggers ingest

$0.005/run

3 Read

Fetch API

Report page → clean markdown/JSON, up to 10 URLs/call, with published_date

free ≤1k URLs/day

4 Extract

Agent API

POST /v1/automation/run-async with goal + output_schema → pathogen, places, cases, deaths, dates

$0.016/step

Our validation gate — before anything reaches the map

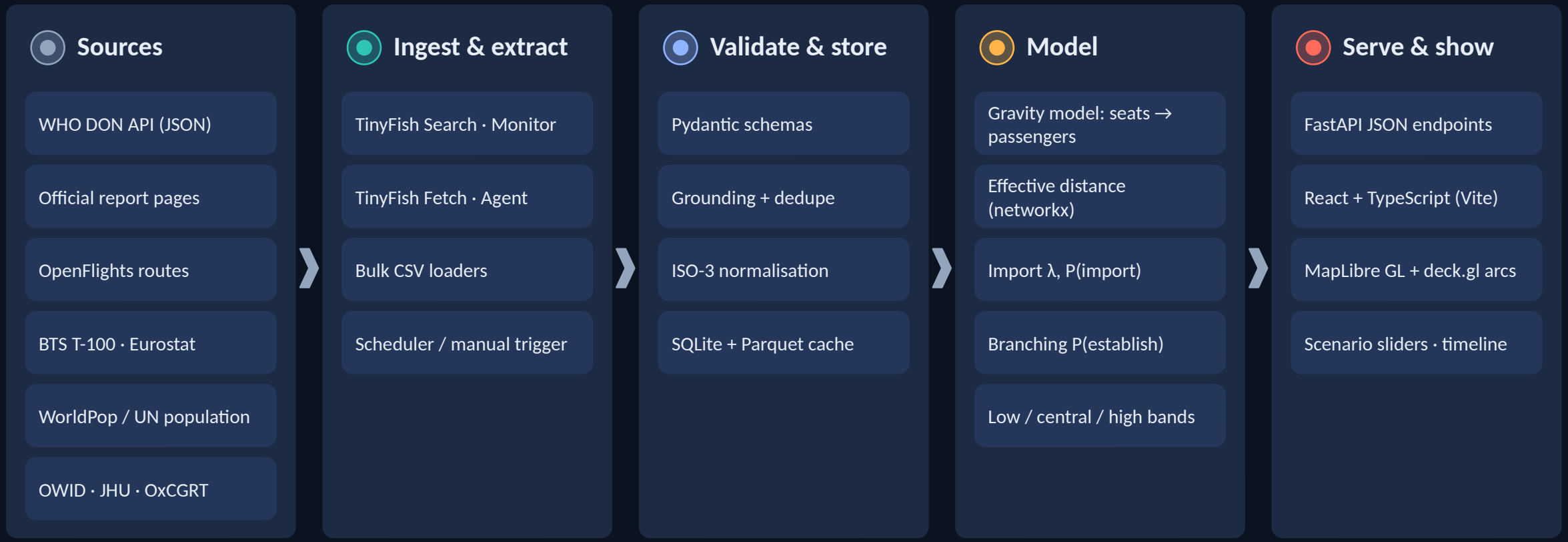
- Grounding check: every extracted number must appear verbatim in the Fetch text, else flagged 'unverified'
- Schema + ISO-3166 country codes, non-negative counts, report date ≤ retrieval date, dedupe by source URL
- Provenance on every record: source URL, run_id, retrieved_at, published_at, status

Agent → border-policy flag σ

The briefing note says live outbreaks need a manual policy flag (OxCGRT ended in 2022). An Agent run reads official entry-screening / quarantine pages into {measure, date, url}; a human confirms it before σ changes.

Server-side only; API key in env vars, never in the browser or repo. Limits per TinyFish docs/pricing (15 Oct 2026): Agent 2 concurrent runs, new wallets start with $8. TinyFish is an extractor — it is not a source of epidemiological truth or passenger data.

Technical architecture and stack



Every value carries a type

observed · estimated · modelled · scenario — the UI colours and labels each differently

Works offline

a cached snapshot of sources lets the demo run if any live API fails

One small service

no microservices or paid data; a laptop runs the whole stack

Calibrate on COVID-19, test on outbreaks it never saw

Input	Source	Honest caveat
Live outbreaks	WHO DON API + TinyFish-extracted pages	Narrative counts; reporting lags
Passengers F	BTS T-100 & Eurostat (observed); gravity model elsewhere	No free global O–D bookings → labelled proxy
Route topology	OpenFlights; OpenSky flight counts 2019–22	Counts flights, not people
Prevalence I/N	OWID / JHU cases ÷ WorldPop or UN population	Testing bias → under-reporting band
Policy σ	OxCGRT (to 2022); TinyFish flag for live	Needs human confirmation

Better labels than ‘first confirmed case’

First official case dates mostly measure who was testing. Where available we use Nextstrain lineage-introduction dates as the arrival label, and we report rank correlation per outbreak — one pandemic cannot prove universal accuracy.

Validation plan (to be run — no accuracy claimed yet)

Fit — COVID-19 first wave: arrival time vs effective distance; fit the passenger coefficient, then freeze it

Test — Delta & Omicron introduction order; H1N1 2009 and SARS 2003 rank correlation, nothing refit

Control — MERS & Ebola: low α must keep international establishment low — if it fires, α is wrong

Contrast — Zika: high importation, establishment set by mosquito climate, not flights

Sources: OWID covid-19-data; JHU CSSE archive; OxCGRT final release; BTS T-100; Eurostat avia; OpenFlights; OpenSky COVID dataset (Zenodo); WorldPop (Huang 2013, Mao 2015); Nextstrain. Plan from the 14 Oct 2026 briefing note.

Scalability, feasibility and use case

P0 · demo in 24 h

- Map with real WHO DON records + source links
- TinyFish Fetch→Agent extraction with validation gate
- Import model with low/central/high bands
- Country panel: presence vs import vs establish
- Offline cached snapshot

P1 · if core works

- Hub arcs + effective-distance ranking
- Gravity-model passenger proxy
- TinyFish Monitor webhook ingest
- σ policy flags with human confirm

P2 · after the hackathon

- Full COVID calibration + held-out tests
- Seats × load factor from live schedules
- Sub-national (admin-1) resolution
- Licensed O–D bookings if available

Use case walk-through

WHO publishes a new outbreak report → Monitor fires → Fetch + Agent extract it → it appears on the map within minutes with its source → an airport health officer selects India, sees the 14-day importation range and the top three origin routes, and drags σ to compare screening options.

Why it is feasible

Free — every dataset is free; paid feeds are optional upgrades

$8 — starter wallet ≈ 500 Agent steps; TinyFish Search and Fetch are free

Offline — cached snapshot keeps the demo alive if APIs fail

How it scales

Countries are nodes and routes are edges (~200 × 200), so recomputing every destination is a sub-second matrix job; new sources are a new extraction goal + schema, not new code.

Honest by design: safeguards, scenario timeline, roadmap

Safeguards built into the product

- Observed, estimated, modelled and scenario values are labelled and coloured differently
- No number without inputs: missing data shows ‘not estimable’, never a guess
- Ranges, not point values; every assumption is a visible slider
- Official WHO / national travel guidance stays the action layer; we link to it

Roadmap

Finale (24 h)	P0 demo with live WHO data and TinyFish ingest
+1 month	COVID calibration and held-out H1N1 / SARS report, published openly
+3 months	Alerts for subscribed destinations; regional sources via new TinyFish goals

Scenario risk-threshold timeline (replaces ‘last safe day to leave’)



Upper-band inputs (high under-reporting, high R, σ = 0, schedules unchanged) with a user-set threshold. It is a scenario marker, not a safety date; leaving late can export infection to others. Follow WHO and national advisories.

References

- Brockmann & Helbing. Science 2013;342:1337–42
- Lloyd-Smith et al. Nature 2005;438:355–9
- Huang et al. PLOS ONE 2013;8:e64317 · Mao et al. J Transp Geogr 2015;48:52–60
- WHO DON API: who.int/api/news/diseaseoutbreaknews
- OWID covid-19-data · JHU CSSE · OxCGRT · BTS T-100 · Eurostat · OpenFlights · OpenSky · Nextstrain · GLEAM
- TinyFish docs: docs.tinyfish.ai (Agent, Fetch, Search, Monitor), accessed 15 Oct 2026